

Product Requirement Document (PRD)

AI Customer Feedback Synthesis & Opportunity Prioritization Engine

Document Status	Approved / v1.0 MVP	Target Audience	Product, Engineering, Executive Leadership
Author	AI Product Manager	Created Date	August 2026
Core Tech Stack	Python 3.12, OpenAI GPT-4o (Structured Outputs), Pandas, Streamlit		

1. Executive Summary & Problem Statement

The Problem: Modern SaaS product teams receive hundreds of raw feedback items across sales calls, support tickets, CSAT surveys, and customer interviews. Synthesizing this qualitative text into actionable feature opportunities takes product managers hours or days each week. Furthermore, manual prioritization often lacks financial objectivity, resulting in low-impact requests competing against critical high-ARR churn risks.

The Solution: The AI Opportunity Prioritization Engine automates feedback clustering, extracts qualitative pain points via LLM structured parsing, enriches feedback with revenue metadata, and computes an objective priority score aligned with company OKRs.

2. Core Objectives & Key Metrics

- Primary Goal:** Reduce feedback synthesis and prioritization time from days to under 10 seconds.
- Data-Driven Prioritization:** Eliminate guesswork by factoring ARR (Annual Recurring Revenue) impact into opportunity rankings.
- Strategic Alignment:** Dynamically re-rank feature opportunities based on custom quarterly company OKRs.

Success Metric	Baseline (Manual Process)	Target Goal (AI Engine)
Feedback Processing Time	5–10 hours / sprint	< 15 seconds
Revenue Visibility	Ad-hoc / Qualitative guessing	100% ARR attribution per cluster
OKR Alignment Accuracy	Subjective PM bias	Algorithmic semantic match (0.0 to 1.0)

3. User Persona & Use Cases

Primary Persona: Product Manager (SaaS)

- Pain Points:** Drowning in customer complaints; struggling to justify roadmap priorities to executives with financial data.
- Needs:** A fast, automated tool to group customer requests, calculate business impact, and export a clear priority matrix for stakeholders.

4. Functional Requirements & Feature Specifications

4.1 Automated Problem Clustering (LLM Intelligence)

The system ingests raw CSV feedback text and uses OpenAI GPT-4o with strict Pydantic schema validation to extract normalized problem clusters. Each cluster includes:

- **Cluster Title & ID:** Concise problem naming (e.g., Enterprise SSO & Authentication).
- **Urgency Score (1–10):** AI-evaluated pain intensity based on customer language severity.
- **Strategic OKR Alignment (0.0–1.0):** Semantic evaluation against active company objectives.
- **Estimated Effort (1–5):** Qualitative engineering effort score.
- **Executive Summary:** 2-sentence synthesis of customer pain points.

4.2 Revenue Metadata Enrichment (Data Pipeline)

The Python pipeline filters the underlying CRM dataset using mapped feedback_id arrays to sum total impacted ARR for each cluster without LLM math hallucinations.

4.3 Algorithmic Prioritization Scoring Engine

Each cluster receives an objective **Priority Score** calculated deterministically using the formula:

```
ARR Multiplier = 1 + (Total Impacted ARR / 100,000)
Raw Impact = (Volume × Urgency Score × OKR Alignment) × ARR Multiplier
Priority Score = Raw Impact / Estimated Effort
```

4.4 Interactive Web UI (Streamlit Dashboard)

- Drag-and-drop CSV uploader for custom datasets.
- Dynamic text area to modify company OKRs in real time.
- Interactive sortable results table showing priority rankings, ARR, and summaries.
- One-click CSV download of the generated roadmap matrix.

5. System Architecture & Data Flow

Data Flow Pipeline:

```
Raw CSV Input → GPT-4o Structured Parser → Pandas Metadata Enricher → Priority Scoring Engine
→ Streamlit Dashboard Display
```

6. Non-Functional Requirements

- **Reliability:** API error handling for rate limits or missing credentials.
- **Data Privacy:** No customer data retained by external providers beyond API session execution.
- **Usability:** Zero-code setup for end users via local web UI.